

REFRIGERATION ENGINEER'S REPORT

Cold Chain Mechanical Group · Our reference CCMG-2026-0114

INSTRUCTIONS

Instructed by:	Talbot & Rennick Insurance Brokers, Inc.
On behalf of:	Harborline Cold Storage & Logistics, LLC
Policy number:	CP-4471-88210
Author:	Ing. Marguerite Delacroix-Iwu, PE
Instructed by:	Talbot & Rennick Insurance Brokers, Inc.
Instructed on:	11 January 2026
Site attendance:	11 and 12 January 2026
Scope:	Cause of the header failure and the extent of plant damage

AFFECTED ROOMS AND PLANT

Room 4 — freezer:	Customer product, ammonia contamination
Room 5 — freezer:	Customer product, ceiling panel damage
Room 6 — freezer:	Customer product, ammonia contamination
Engine room:	Water damage, 3,600 sq ft
Electrical room:	Two compressor VFD cabinets wetted

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Circumstances and findings

CIRCUMSTANCES

Circumstances:

The plant is a two-stage anhydrous ammonia system serving five freezer rooms and a dock. The failure occurred at a welded elbow in the ceiling space of Room 5, on a header installed as part of an evaporator retrofit completed in 2023. Detection alarmed at 25 ppm and the plant tripped at 150 ppm, both as designed.

FINDINGS

Findings:

The failed elbow shows a circumferential crack initiating at the weld toe on the inside radius. The weld cap is undercut over approximately 40% of its circumference and the root pass is incompletely fused across a 22 mm arc. The failure is consistent with fatigue propagating from a weld defect under normal pressure cycling, not with overpressure. The section has been cut out, tagged and retained. The protective safeguards required by the policy — detection, automatic shutdown and room temperature alarms — were all in service and functioned; the release was limited by them rather than worsened.

Date of loss:	10 January 2026
Cause of loss:	Ammonia release — weld failure on a liquid ammonia header
Incident reference:	BCFD-2026-001188
Injuries reported:	0

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Preliminary quantum and reserve advice

PRELIMINARY QUANTUM

Basis:

The figures below are preliminary and are given for reserving purposes only. They are subject to verification and may move as the investigation continues.

Ammonia piping repair and re-commissioning:	USD 186,000
Two compressor VFD cabinets:	USD 148,000
Evaporator coil cleaning and re-certification:	USD 86,000
Engine room and electrical room reinstatement:	USD 310,000
Room 5 insulated panel ceiling:	USD 120,000
Estimated loss:	3,700,000
Currency:	USD

COMMENT ON THE 2023 RETROFIT

Comment:

The retrofit was carried out by Northern Bay Refrigeration Services. The weld in question is theirs. Their weld procedure qualification records and the radiographic inspection reports for that scope should be preserved; on the evidence available a recovery against that contractor is worth pursuing.

RECOMMENDATION

Recommendation:

Do not return the affected rooms to service until the replacement spool has been radiographed and the system has held a 24-hour pressure test.